Project Report

EE 517 Fall 2024

HW 4 — RCS

Justin Pulley

For: Prof. K Russo

Date Performed: 4/9/2024

In this project, I modeled 10 complex targets as a set of 50 point-scatterers that are randomly distributed in a box that is ±5 meters in the x dimension (along the initial radar boresight) and ±2.5 meters in the y dimension, centered at (x,y) = (0,0). Scatterers are located at (x,y) coordinate pairs.

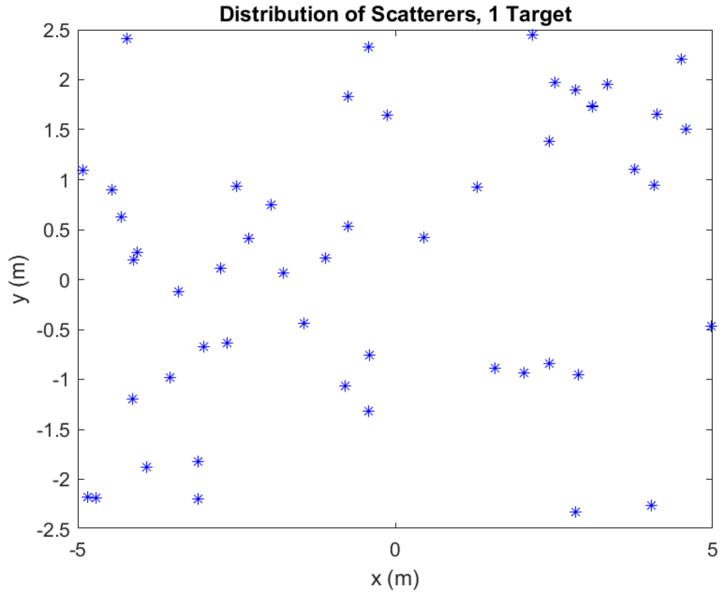


Figure 1. The blue asterisks show the distribution of 50 scatterers of fixed RCS = 1 m^2 each for 1 target.

I simulated a monostatic radar, with a transmit frequency of 10 GHz, on an airborne vehicle orbiting a target located 10km away from the center of the target. The total RCS of the target is calculated at each aspect angle from 0° to 359° in 0.5° intervals, by summing each relative RCS of each scatters at each angle. The db RCS value was calculated and plotted against the aspect angle. The mean of

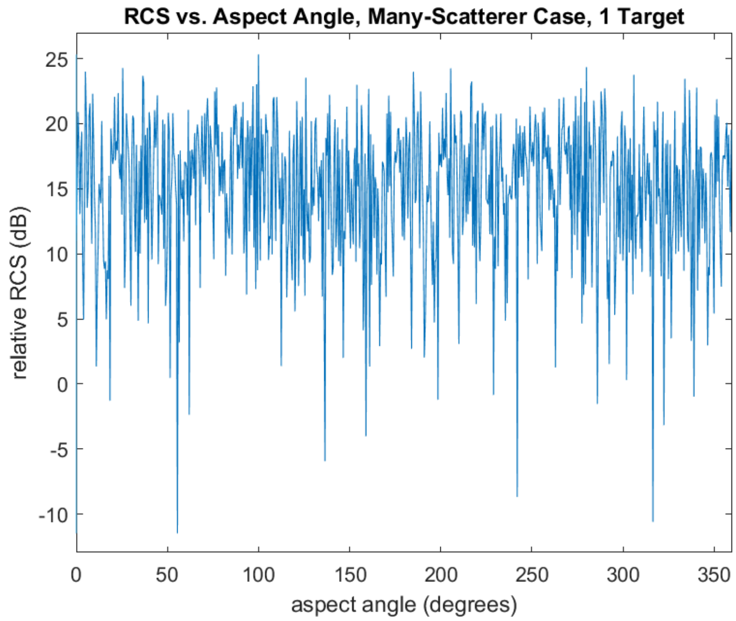


Figure 2. Relative RCS of a complex target vs. aspect angle at a range of 10 km and transmit frequency of 10 GHz.

The RCS fluctuations shown in Figure 2 make the use of a statistical description of the RCS more convenient. A specific exponential probability density function (PDF) can model the RCS of the scatters within a single-resolution cell. The PDF used for this project was for the use case of a target that has many scatterers, randomly distributed, and none being dominant, which matched the parameters of the modeled targets. I calculated and plotted this PDF for all complex targets. For the PDF, I had to calculate the mean of the linear scale RCS. In this case, the mean of the RCS values would approximate the individual scatter's RCS as the number of scatters increases. Additionally, I modeled a histogram of RCS data with varying aspect angles for the targets, which matched the PDF. The RCS data for this histogram was normalized.

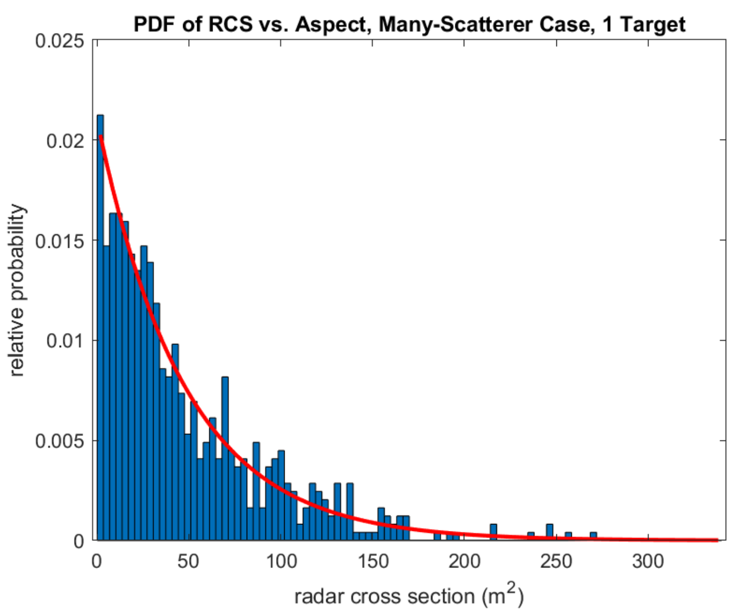


Figure 3. Histogram of RCS data with varying aspect angles and exponential PDF with the same RCS mean for a single target

I computed the average RCS data histogram over 10 different targets. This gave a better approximation of the theoretical exponential PDF than the single target histogram because as the number of samples increases the values tend toward the theoretical values.

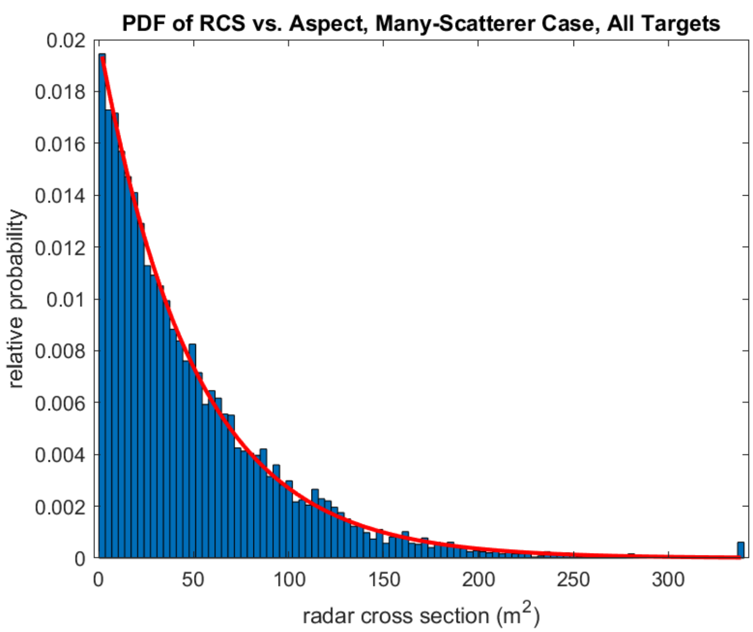


Figure 4. Averaged histogram of the RCS data with varying aspect angles and exponential PDF with the same RCS mean.

For 10 targets, I calculated the RCS for 10 targets at a fixed aspect angle, but varying RF frequency. Just like the aspect angle, a varying RF frequency causes RCS fluctuations. This was modeled in a histogram of normalized RCS values with a varying frequency from 10 GHz to 14.5 GHz in 15 MHz intervals. Next, the RCS data histograms of 10 targets were averaged into one histogram. Along with the theoretical exponential PDF for our specific complex targets. The averaged 10-target histogram better matched the PDF, than the single-target histogram. This is due to the averaged histogram having more sample values which causes the histogram to tend more towards the PDF.

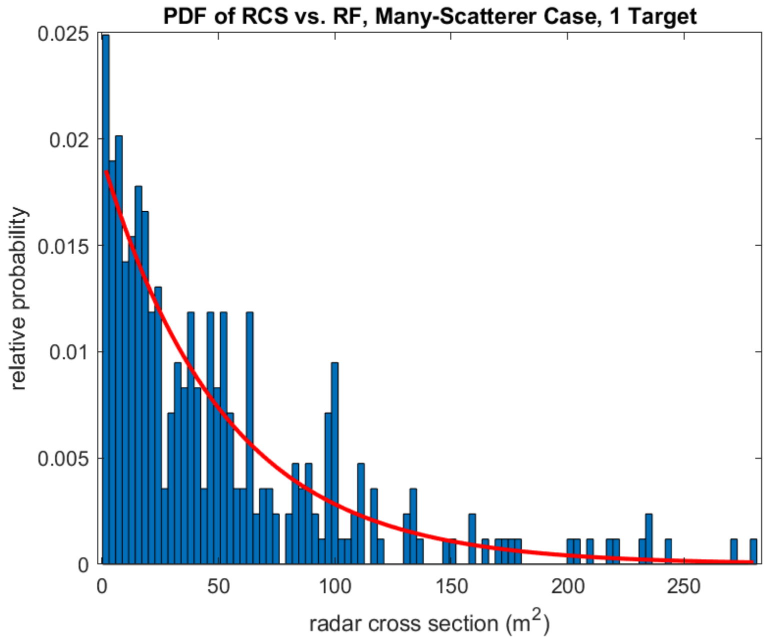


Figure 5. Histogram of RCS data with varying RF frequency and exponential PDF with the same RCS mean for a single target

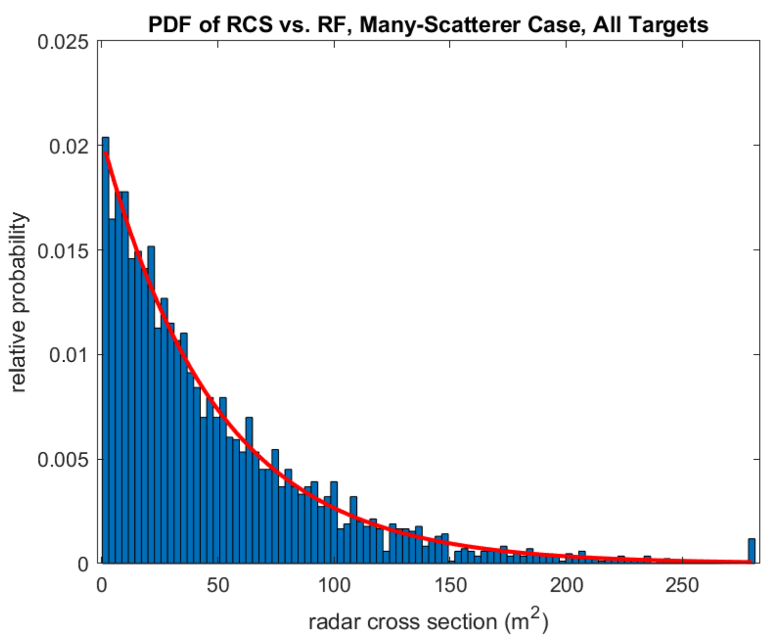


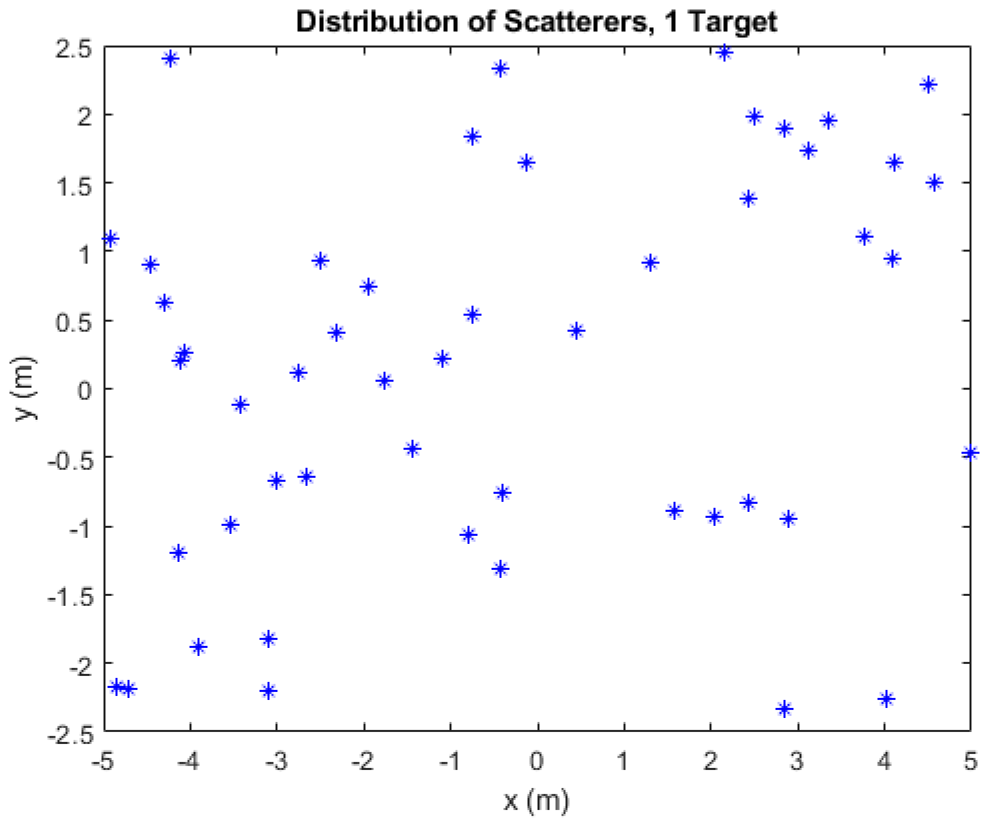
Figure 6. Averaged Histogram of RCS data with varying RF frequency and exponential PDF with the same RCS mean.

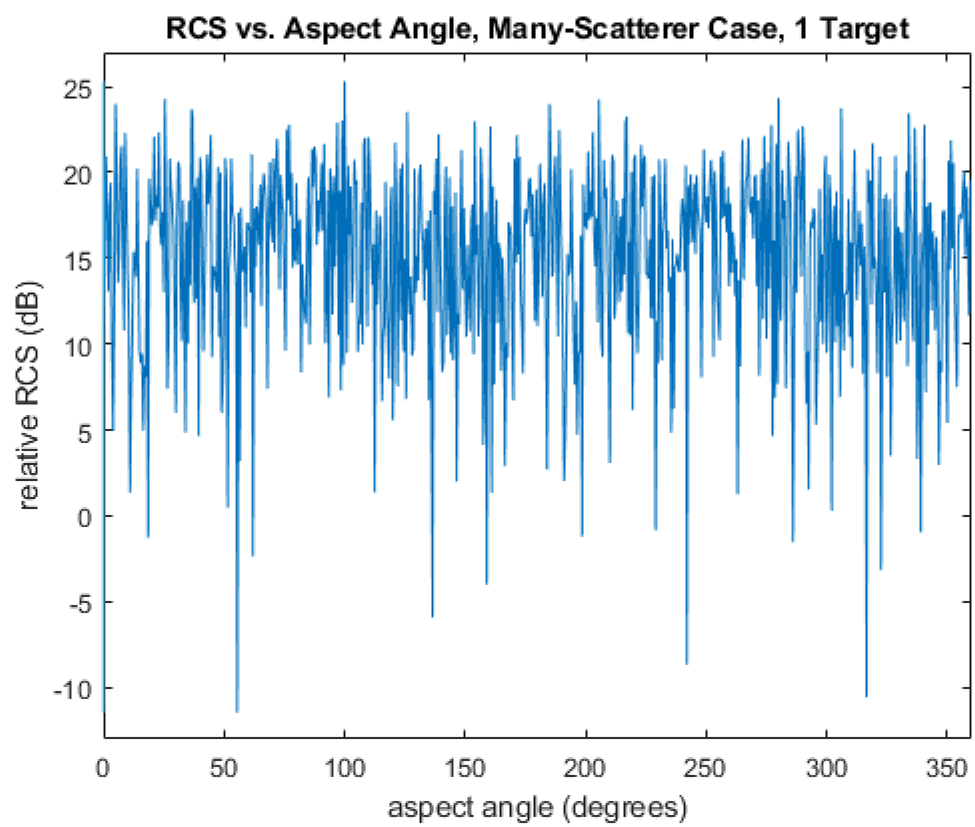
```
close all
clear
clc

Nt = 10; % number of targets
N = 50; %number of scatteres
z = 1; % RCS of each individual scatter sigma_i = 1 m^2

% 50 point scatters uniformly random distributed in a box +- 5 meters, and
% +- 2.5 meters in y dimension centered at (x,y) = (0,0), for 10 different
% targets

figure(1)
x = 10*rand(N,Nt)-5;
y = 5*rand(N,Nt)-2.5;
plot(x(:,1),y(:,1),'*b')
axis([-5 5 -2.5 2.5]);
xlabel('x (m)');
ylabel('y (m)');
title('Distribution of Scatterers, 1 Target');
saveas(gcf,'Distribution of Scatterers, 1 Target.png');

% Input number of angles and nominal range and frequency
% M=input('Enter # of angles: ');
% R=input('Enter nominal range (m): ');
% f=input('Enter frequency (Hz): ');
M = round(360/0.5);
R = 10e3;
f = 10e9;

% Loop over aspect angle to do complex voltage estimates
q = zeros(M);
echo=zeros(M,Nt);

% Loop over radar-target aspect angles

for k=1:M
    q(k)=(2*pi/M)*(k-1); % current aspect angle

    % Loop over targets and individual point scatterers
    % each column of 'echo' is a different target

    for r = 1:Nt
        for p=1:N
            phasor=z*exp(j*4*pi*f*norm([x(p,r)-R*cos(q(k)),y(p,r)-
R*sin(q(k))])/3e8);
            echo(k,r)=echo(k,r)+phasor;
        end
    end
end

% The magnitude-squared of the total complex echo
```

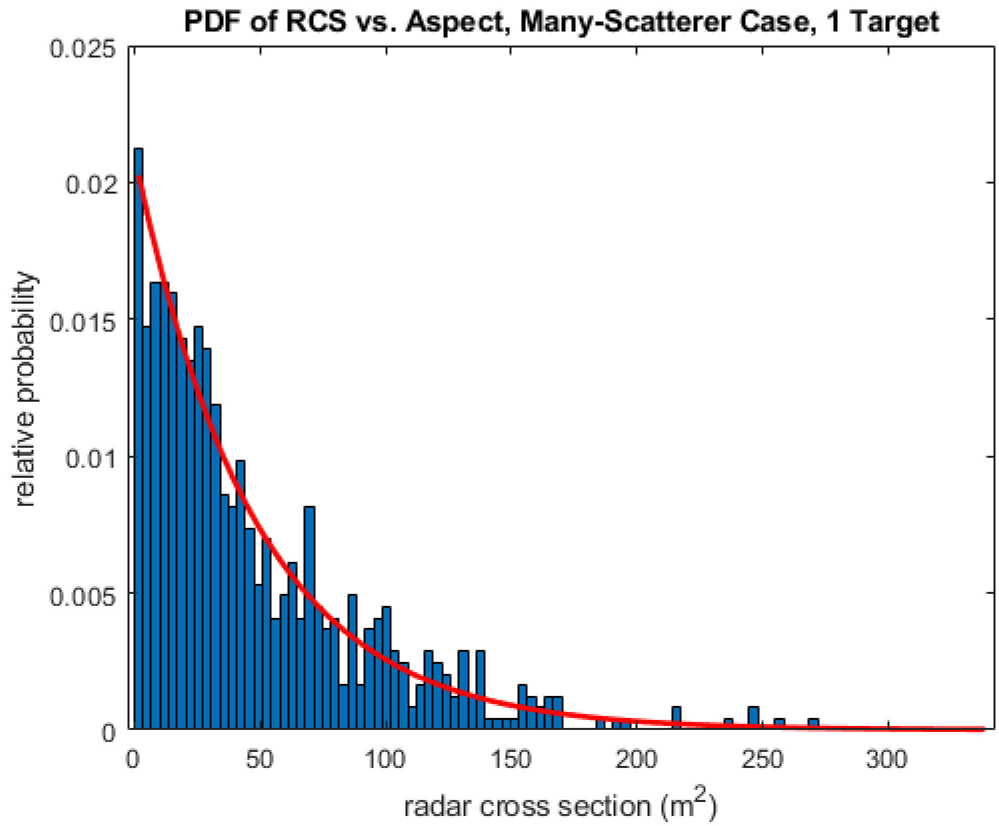
```
% is the RCS to within a constant

RCS = abs(echo).^2;
RCSdB = 10*log10(RCS);

% Plot the dB data for the first target

figure(2)
plot((180/pi)*q,RCSdB(:,1));
xlabel('aspect angle (degrees)');
ylabel('relative RCS (dB)');
title('RCS vs. Aspect Angle, Many-Scatterer Case, 1 Target')
axis([0,360,10*log10(N*z)-30,10*log10(N*z)+10]);
saveas(gcf,'RCS vs. Aspect Angle, Many-Scatterer Case, 1 Target.png');

% Compute a 100-bin histogram of the data for each target;
% plot the theoretical exponential distribution as well

density = zeros(100,Nt);
centers = [];
for r = 1:Nt
    if (isempty(centers))
        [count,centers]=hist(RCS(:,r),100);
    else
        [count,centers]=hist(RCS(:,r),centers);
    end
    bin_size = centers(2) - centers(1);
    area = sum(count)*bin_size;
    density(:,r) = count/area;
    figure(3)
    bar(centers,density(:,r),1);
    hold on
    msig = mean(RCS(:,r)); % compute mean of current linear-scale RCS data
    exp_pdf = exp(-centers/msig)/msig; % theoretical pdf on same x-axis values

    plot(centers,exp_pdf,'r','LineWidth',2)
    xlabel('radar cross section (m^2)')
    ylabel('relative probability')
    title('PDF of RCS vs. Aspect, Many-Scatterer Case, 1 Target')
    hold off
    pause(0.5)
end
saveas(gcf,'PDF of RCS vs. Aspect, Many-Scatterer Case, 1 Target.png');

% Now average the separate histograms and plot with an overlaid exponential
% based on the mean of all the data for all of the targets

figure(4);
bar(centers,mean(density,2),1);
hold on
msig = mean(RCS(:));  % compute mean of all linear-scale RCS data
exp_pdf = exp(-centers/msig)/msig;  % theoretical pdf on same x-axis values
plot(centers,exp_pdf,'r','LineWidth',2)
xlabel('radar cross section (m^2)')
```

```
ylabel('relative probability')
title('PDF of RCS vs. Aspect, Many-Scatterer Case, All Targets')
hold off
saveas(gcf,'PDF of RCS vs. Aspect, Many-Scatterer Case, All Targets.png');

% Now do the variation with frequency

% Input number of angles and nominal range and frequency
% Mf=input('Enter # of frequencies: ');
% f1=input('Enter start frequency (Hz): ');
% df=input('Enter freq step (Hz)');

Mf = 300;
f1 = 10e9;
df = 15e6;
fv = f1+(0:Mf-1)*df;

% Loop over radar frequencies

echof=zeros(Mf,Nt);
for k=1:Mf
    % Loop over targets and individual point scatterers
    % each column of 'echof' is a different target
    for r = 1:Nt
        for p=1:N
            phasor=z*exp(j*4*pi*fv(k)*norm([x(p,r)-R,y(p,r)])/3e8);
            echof(k,r)=echof(k,r)+phasor;
        end
    end
end

% The magnitude-squared of the total complex echo
% is the RCS to within a constant

RCSf=abs(echof).^2;
RCSfdB=10*log10(RCSf);

% Plot the dB data for the first target as an example

figure(5)
plot(fv,RCSfdB(:,1)); xlabel('RF frequency (Hz)');
ylabel('relative RCS (dB)');
title('RCS vs. RF Frequency, Many-Scatterer Case, 1 Target')
axis([10e9,14.5e9,10*log10(N*z)-30,10*log10(N*z)+10]);
saveas(gcf,'RCS vs. RF Frequency, Many-Scatterer Case, 1 Target.png');


% Compute a 100-bin histogram of the data for each target;
% plot the theoretical exponential distribution as well

densityf = zeros(100,Nt);
centersf = [];
for r = 1:Nt
    if (isempty(centersf))
```

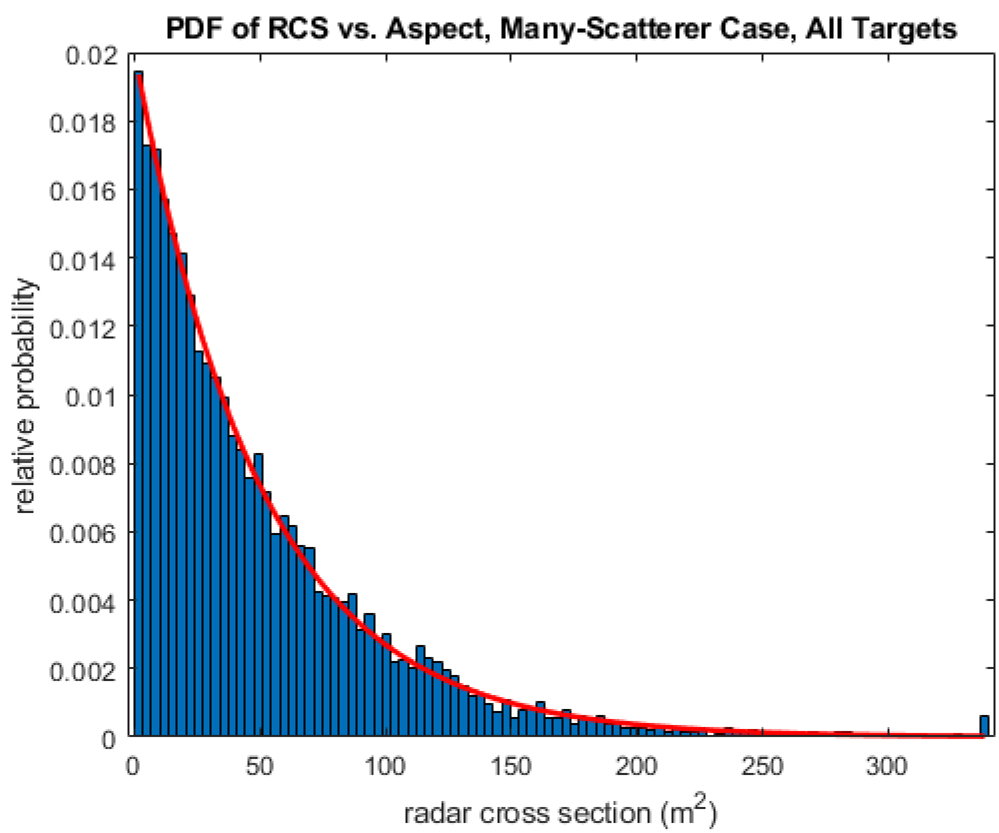
```
        [countf,centersf]=hist(RCSf(:,r),100);
    else
        [countf,centersf]=hist(RCSf(:,r),centersf);
    end
    bin_sizef = centersf(2)-centersf(1);
    areaf = sum(countf)*bin_sizef;
    densityf(:,r) = countf/areaf;
    % Plot the histogram of the data for a single target, and an overlaid
    % exponential for that target, based only on that target's data
        figure(6);
        bar(centersf,densityf(:,r),1);
        hold on
        msigf=mean(RCSf(:,r)); % compute mean of current linear-scale RCS data
        exp_pdf = exp(-centersf/msigf)/msigf; % theoretical pdf on same x-axis
 values

        plot(centersf,exp_pdf,'r','LineWidth',2)
        xlabel('radar cross section (m^2)')
        ylabel('relative probability')
        title('PDF of RCS vs. RF, Many-Scatterer Case, 1 Target')
        hold off
        pause(0.5)
end
saveas(gcf,'PDF of RCS vs. RF, Many-Scatterer Case, 1 Target.png');

% Now average the separate histograms and plot with an overlaid exponential
% based on the mean of all the data for all of the targets

figure(7);
bar(centersf,mean(densityf,2),1);
hold on
msigf=mean(RCSf(:));  % compute mean of all linear-scale RCS data
exp_pdf = exp(-centersf/msigf)/msigf;  % theoretical pdf on same x-axis values
plot(centersf,exp_pdf,'r','LineWidth',2)
xlabel('radar cross section (m^2)')
ylabel('relative probability')
title('PDF of RCS vs. RF, Many-Scatterer Case, All Targets')
hold off
saveas(gcf,'PDF of RCS vs. RF, Many-Scatterer Case, All Targets.png');
```


Distribution of Scatterers, 1 Target
y (m)
x (m)
2.5
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
-2.5
-5
-4
-3
-2
-1
0
1
2
3
4
5

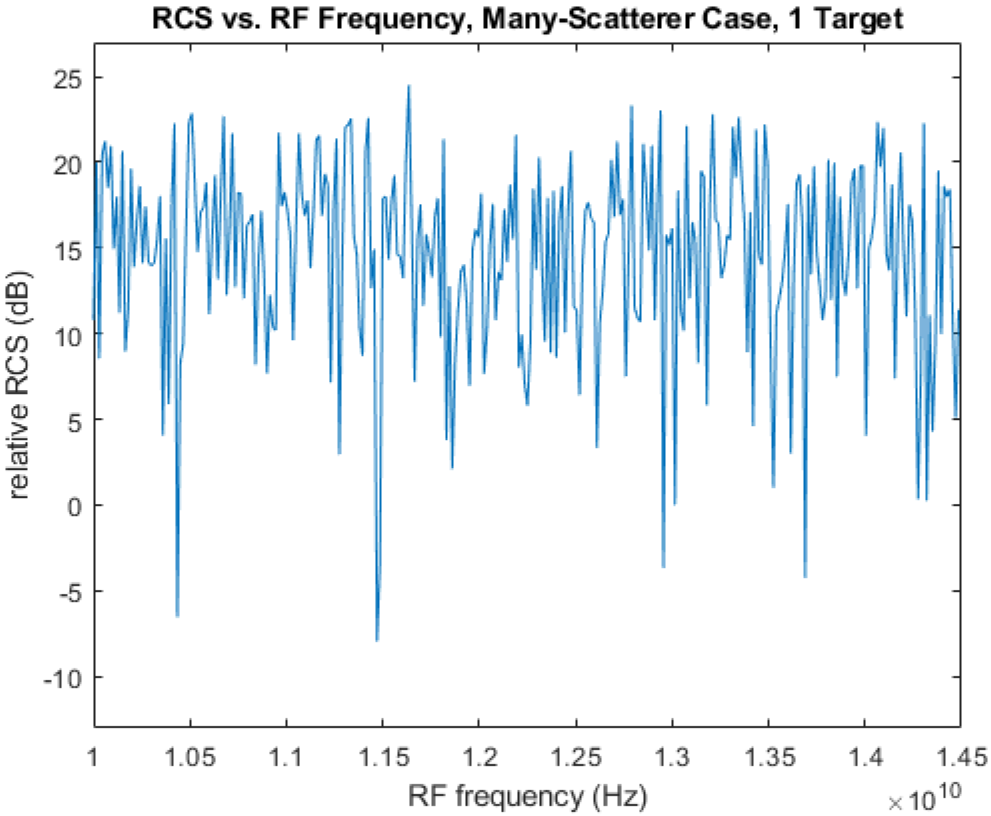

RCS vs. Aspect Angle, Many-Scatterer Case, 1 Target
relative RCS (dB)
aspect angle (degrees)
25
20
15
10
5
0
-5
-10
0
50
100
150
200
250
300
350

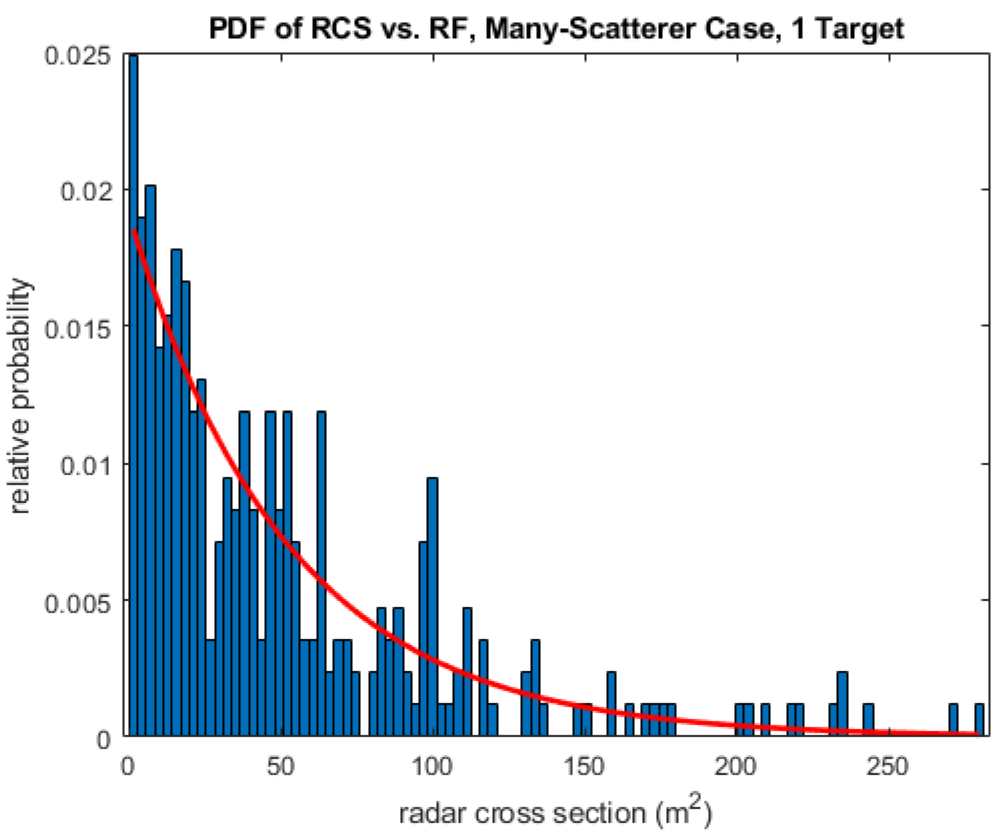


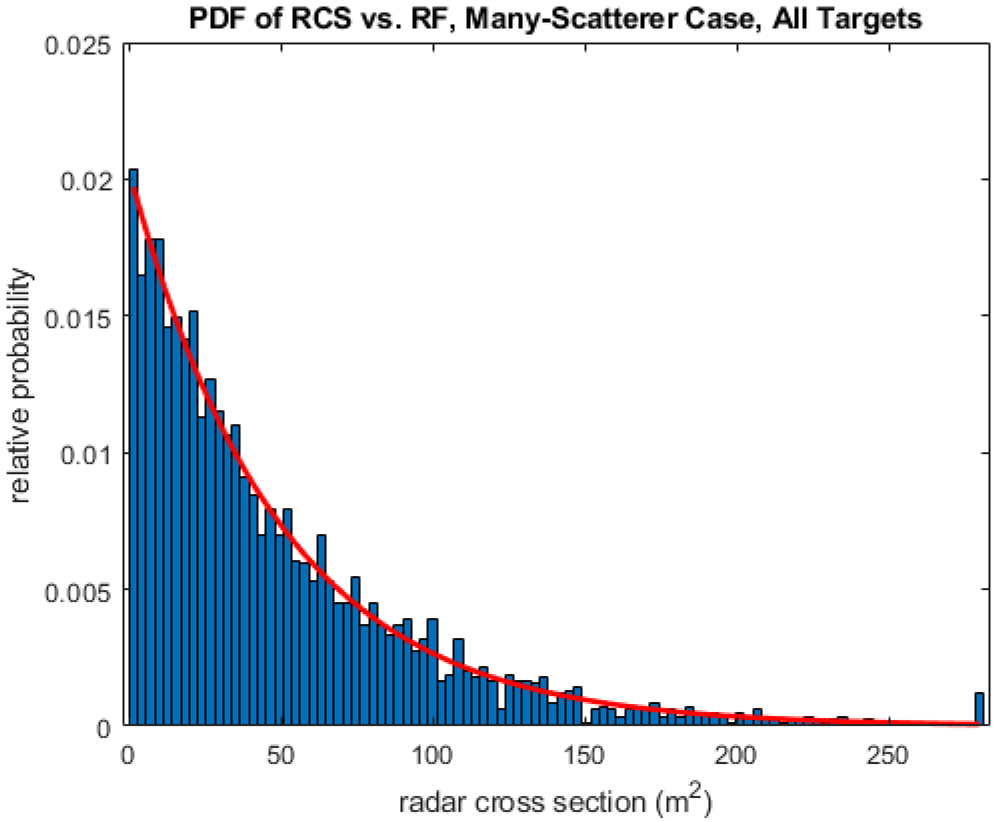
PDF of RCS vs. Aspect, Many-Scatterer Case, 1 Target
relative probability
radar cross section (m2)



PDF of RCS vs. Aspect, Many-Scatterer Case, All Targets
relative probability
radar cross section (m2)


RCS vs. RF Frequency, Many-Scatterer Case, 1 Target
relative RCS (dB)
25
20
15
10
5
0
-5
-10
1
1.05
1.1
1.15
1.2
1.25
1.3
1.35
1.4
1.45
RF frequency (Hz)
$\times 10^{10}$


PDF of RCS vs. RF, Many-Scatterer Case, 1 Target
relative probability
0.025
0.02
0.015
0.01
0.005
0
0
50
100
150
200
250
radar cross section (m^2)



Published with MATLAB® R2022b